Linear Regression Models with R

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Contents

1 Loading Data

We will use the *Palmer Penguin* data set available directly from the `palmerpenguins` package. You can learn more about the data at allisonhorst.github.io/palmerpenguins.

Today, we will explore the variability in penguin weight, starting by looking at **whether size and weight are associated**.

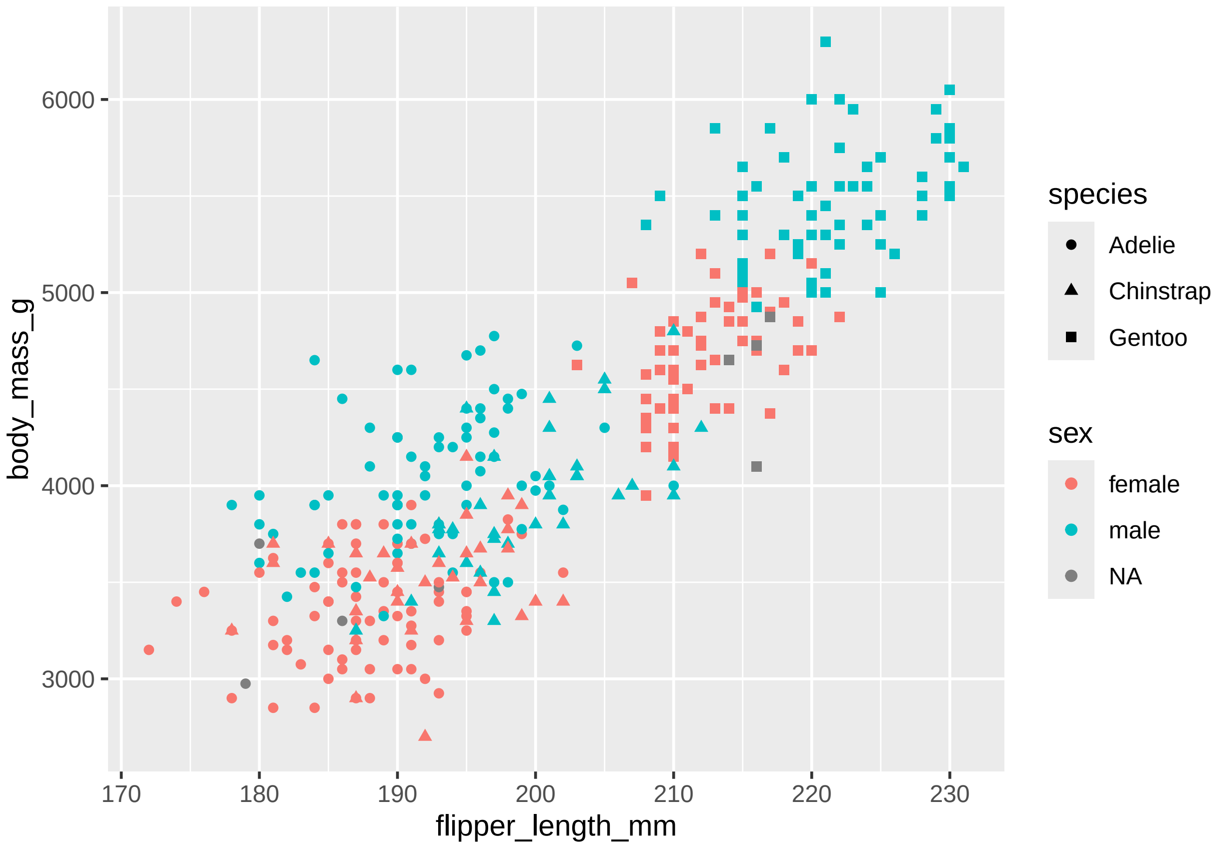
```
# install.packages("palmerpenguins")
library(palmerpenguins)
library(ggplot2)
library(dplyr)
library(car)

dat <- as.data.frame(penguins) # rename dataset
str(dat) # structure of dataset
```

2 Exploring the data

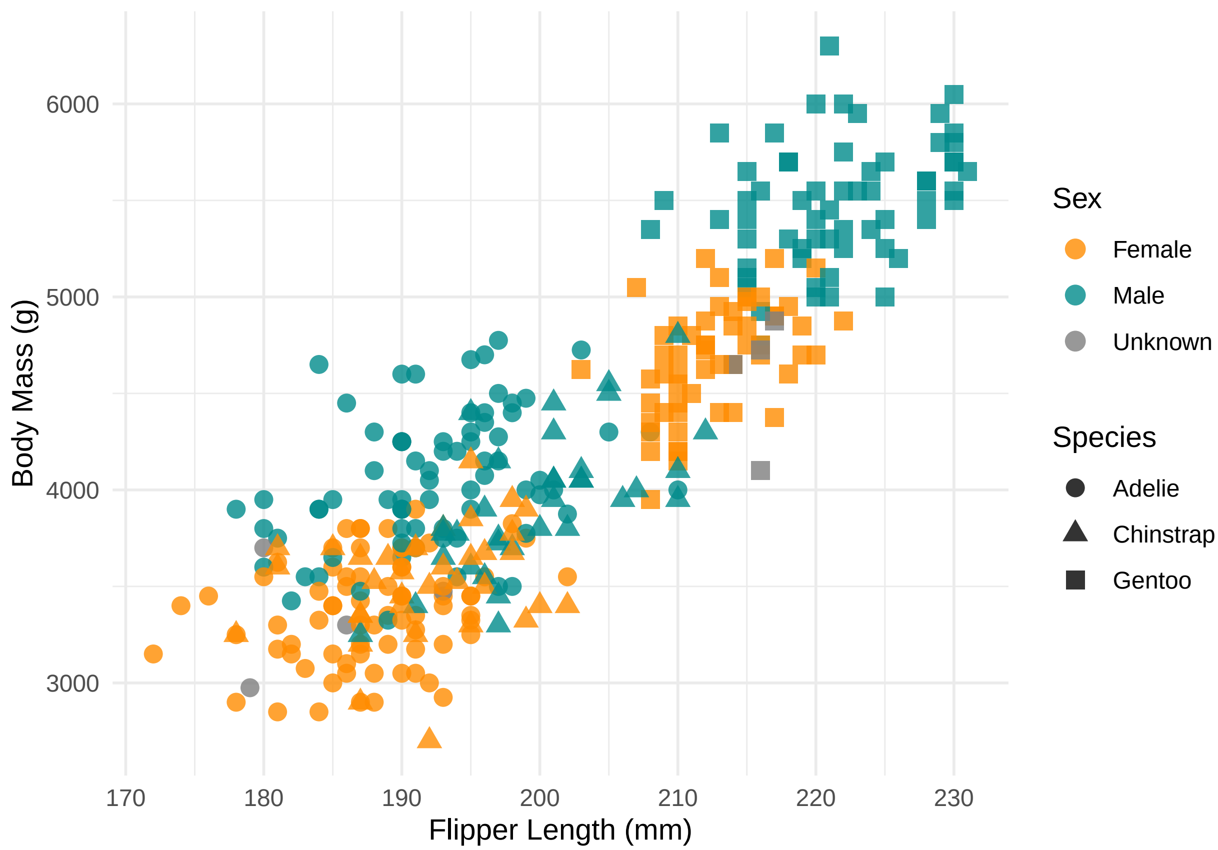
Which one of the variables included in the data may contribute be associated with weight?

2.1 Plot

```
ggplot(dat, aes(x = flipper_length_mm, y = body_mass_g, color = sex, shape = species)) +
  geom_point()
```




```r
# Improve readability

ggplot(dat, aes(x = flipper_length_mm, y = body_mass_g, color = sex, shape = species)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                     name = "Sex",
                     labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
                     name = "Species") +
  labs(
    x = "Flipper Length (mm)",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```



3 Associations Between Continuous Variables

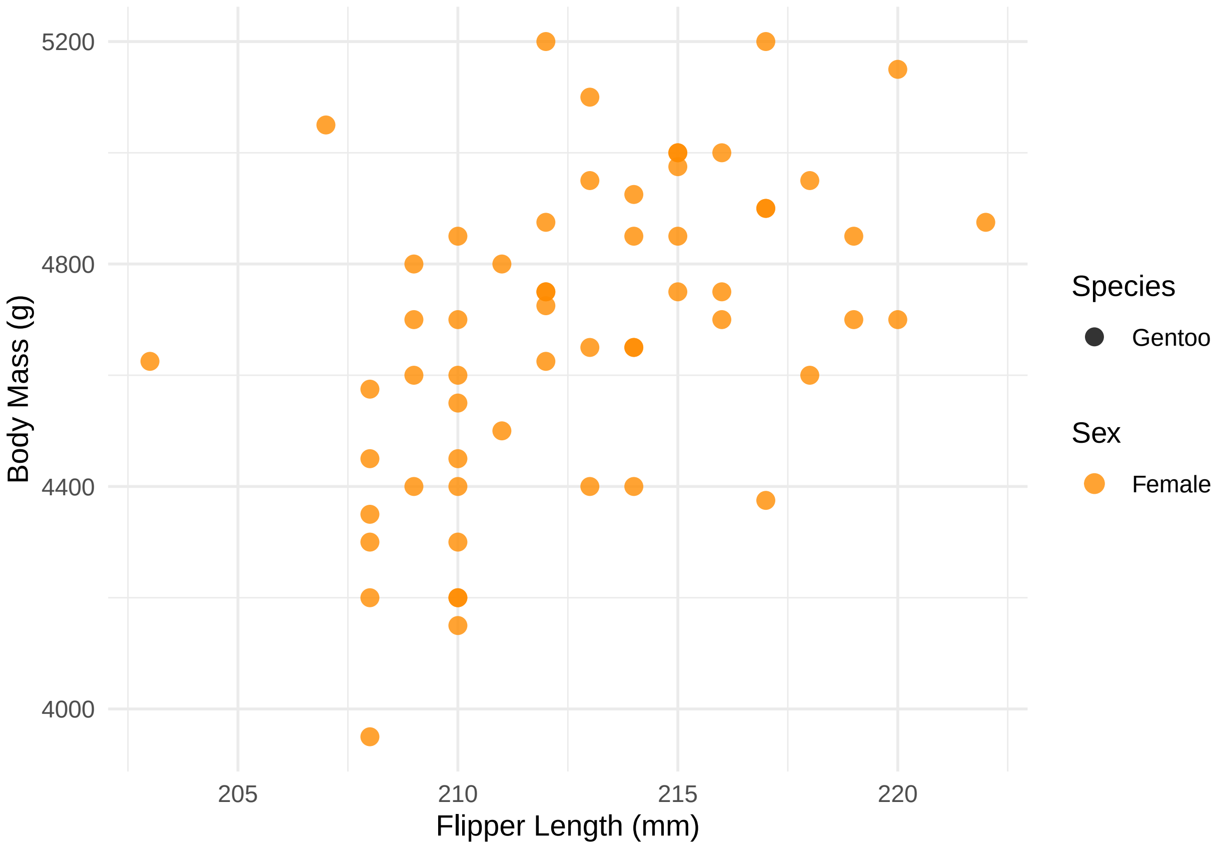
3.1 Subset

Let's focus on one species and one sex so we can study intra-species and intra-sex but inter-individual variability.

```
table(dat$sex, dat$species)

dat_gen_f = subset(dat, species == "Gentoo" & sex == "female")
```

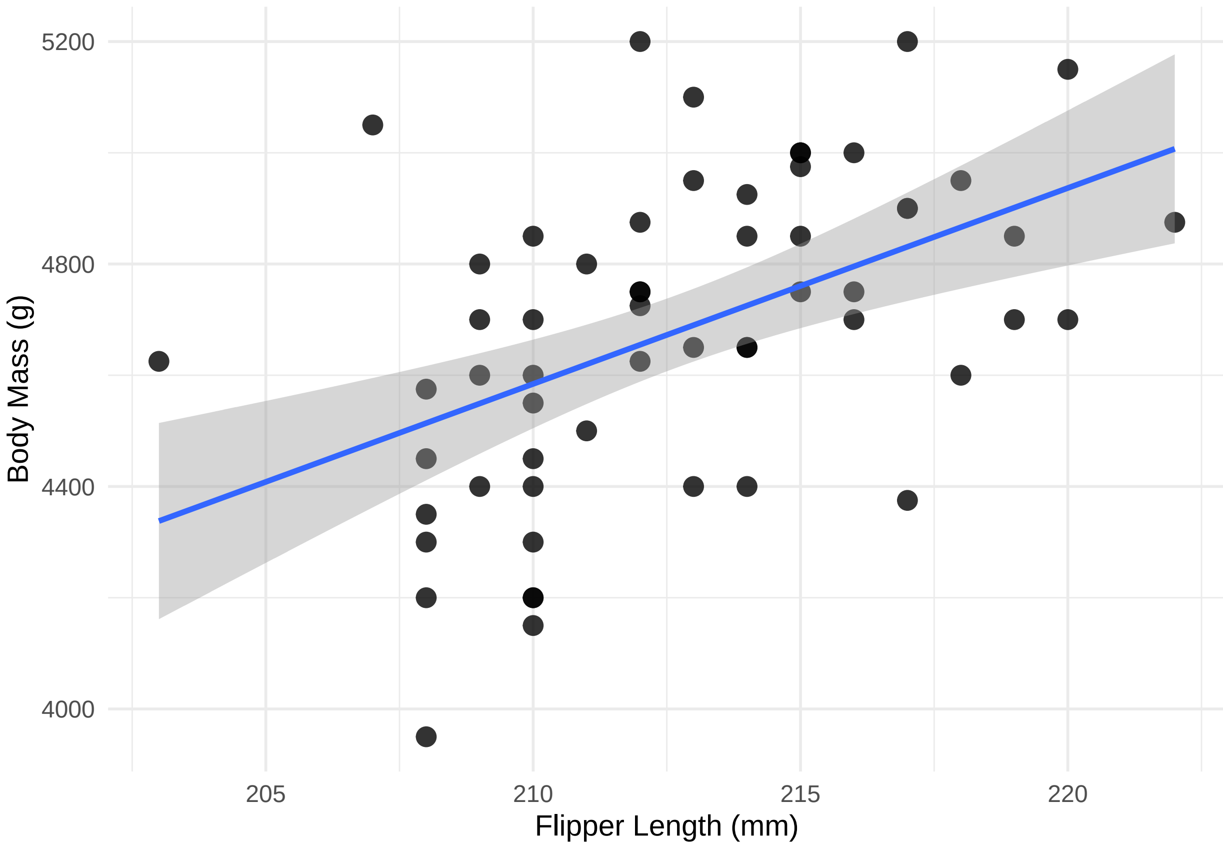
3.2 Re-plot

```
ggplot(dat_gen_f, aes(x = flipper_length_mm, y = body_mass_g, color = sex, shape = species)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                     name = "Sex",
                     labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
                     name = "Species") +
  labs(
    x = "Flipper Length (mm)",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```



Add regression line

```
ggplot(dat_gen_f, aes(x = flipper_length_mm, y = body_mass_g)) +
  geom_point(size = 3, alpha = 0.8) +
  geom_smooth(method = "lm") +
  labs(
    x = "Flipper Length (mm)",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```



Does a linear relationship seem reasonable?

4 Statistical Test

4.1 Assumptions

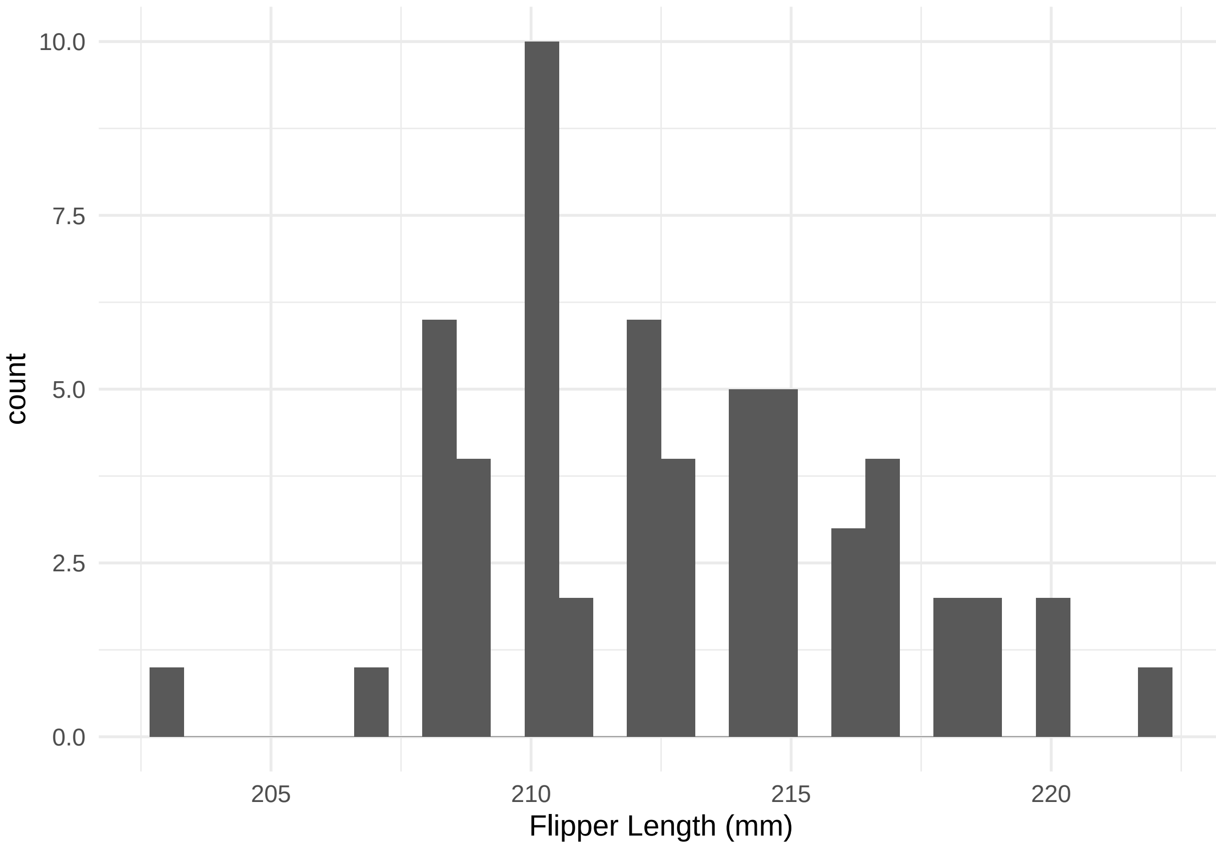
4.1.1 Independance (all tests)

Observations from each population are independent of one another. This is likely the case (would be determined by the study design).

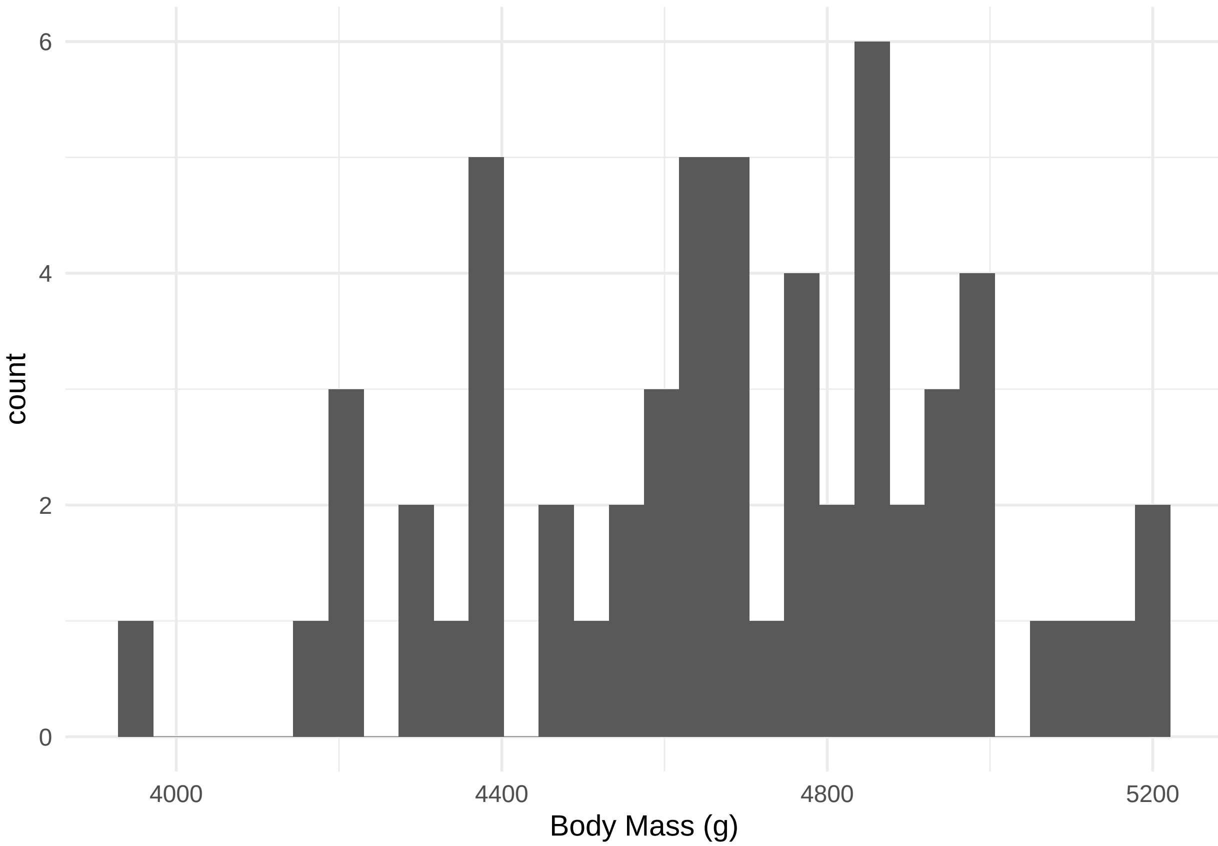
4.1.2 Normality (Pearson and regression)

Pearson - *Response variable is normally distributed for each value of the explanatory variable.* This is easier to assess on the model residuals (after model fitting), although we can already guess on the scatter plot.

```
ggplot(dat_gen_f, aes(x = flipper_length_mm)) +
  geom_histogram() +
  labs(
    x = "Flipper Length (mm)"
  ) +
  theme_minimal()
```




```r
ggplot(dat_gen_f, aes(x = body_mass_g)) +
  geom_histogram() +
  labs(
    x = "Body Mass (g)"
  ) +
  theme_minimal()
```



For a linear regression - Same as above

4.1.3 Homoscedasticity (regression)

Variance of the response variable is equal across each value of the explanatory variable. This is easier to assess on the model residuals (after model fitting), although we can already guess on the scatter plot. .

4.2 Test

- Null hypothesis: there is no association between flipper length and body weight.
- Alternative: there is an association between flipper length and body weight (i.e., variations in individual size contributes to explain variations in individual weight).

4.2.1 Pearson's Correlation Test

Pearson is a parametric test based on the covariance of the two variables.

```
cor.test(dat_gen_f$flipper_length_mm, dat_gen_f$body_mass_g,
         method = c("pearson"))
```

```
##
##  Pearson's product-moment correlation
##
## data:  dat_gen_f$flipper_length_mm and dat_gen_f$body_mass_g
## t = 4.1796, df = 56, p-value = 0.0001035
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2623673 0.6624753
## sample estimates:
```

```
##      cor
## 0.487618
```

Based on Pearson's test, we would reject the null hypothesis. There is a significant association between flipper length and body weight ($p < 0.05$). The Pearson's correlation coefficient [95% confidence interval] of 0.49 [0.26, 0.66], indicates a positive, but weak association (a lot of the variability remains unexplained).

As the normality assumption was not obviously verified, we can check that we reach the same conclusion with a non-parametric test.

4.2.2 Spearman's Rank Test

Spearman is a non-parametric test based on the variables ranks.

```r
cor.test(dat_gen_f$flipper_length_mm, dat_gen_f$body_mass_g,
         method = c("spearman"))
```

```
##
##  Spearman's rank correlation rho
##
## data:  dat_gen_f$flipper_length_mm and dat_gen_f$body_mass_g
## S = 15388, p-value = 2.171e-05
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.5266394
```

Based on Spearman's test, we also reject the null hypothesis. There is a significant association between flipper length and body weight ($p < 0.05$). The Spearman's correlation coefficient of 0.53, indicates a positive, but weak association (a lot of the variability remains unexplained).

Both tests support the same conclusion: our conclusion is robust to the violation of the normality assumption.

Note that this method does not generate an effect size.

4.2.3 Linear Regression Model

Linear models are parametric tests. They can be computed through various methods. A common one is the ordinary least square method.

```r
# Fit model
mod = lm(body_mass_g ~ flipper_length_mm,
          data = dat_gen_f)

# Return outputs (including p-value)
summary(mod)
```

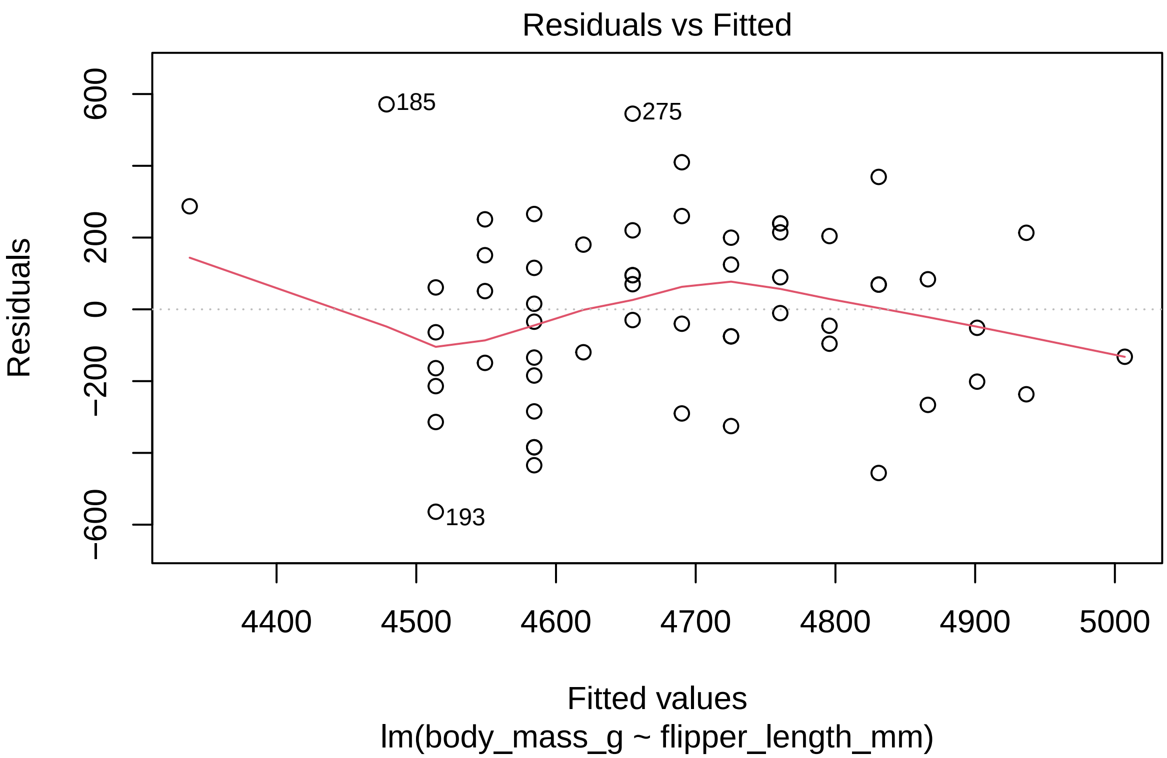
4.2.3.1 Model Fitting

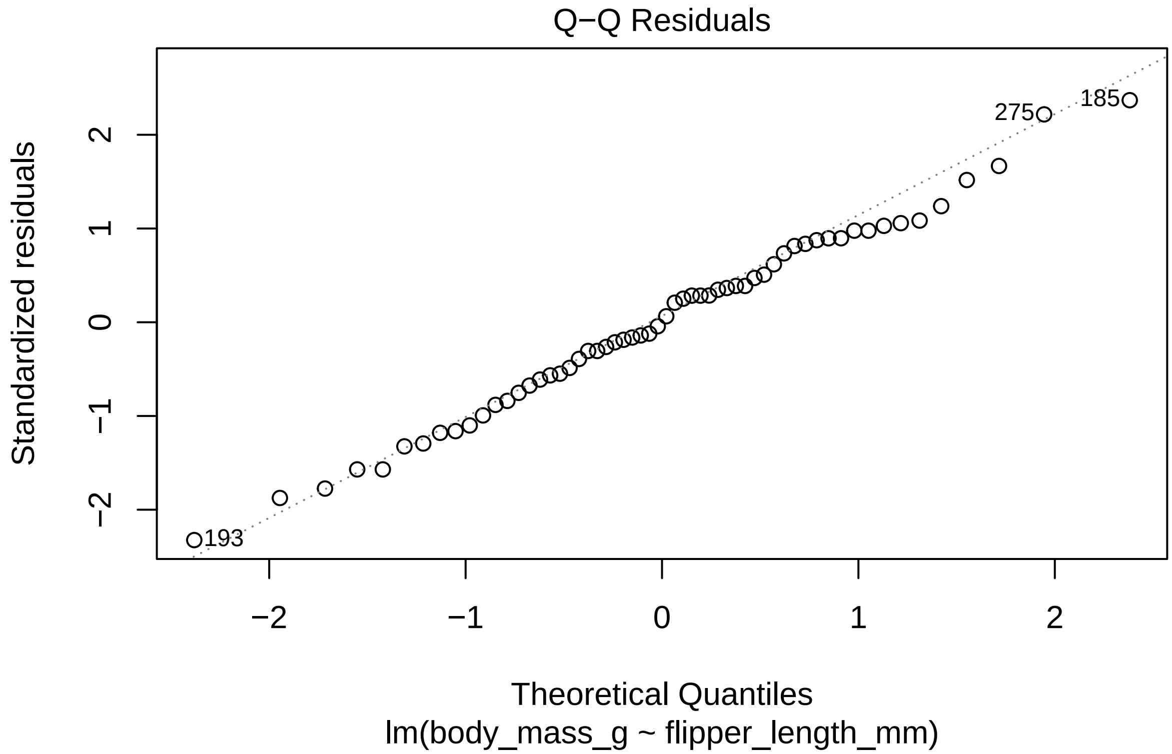
```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm, data = dat_gen_f)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -563.94 -160.25    2.55  194.88  571.29
##
```

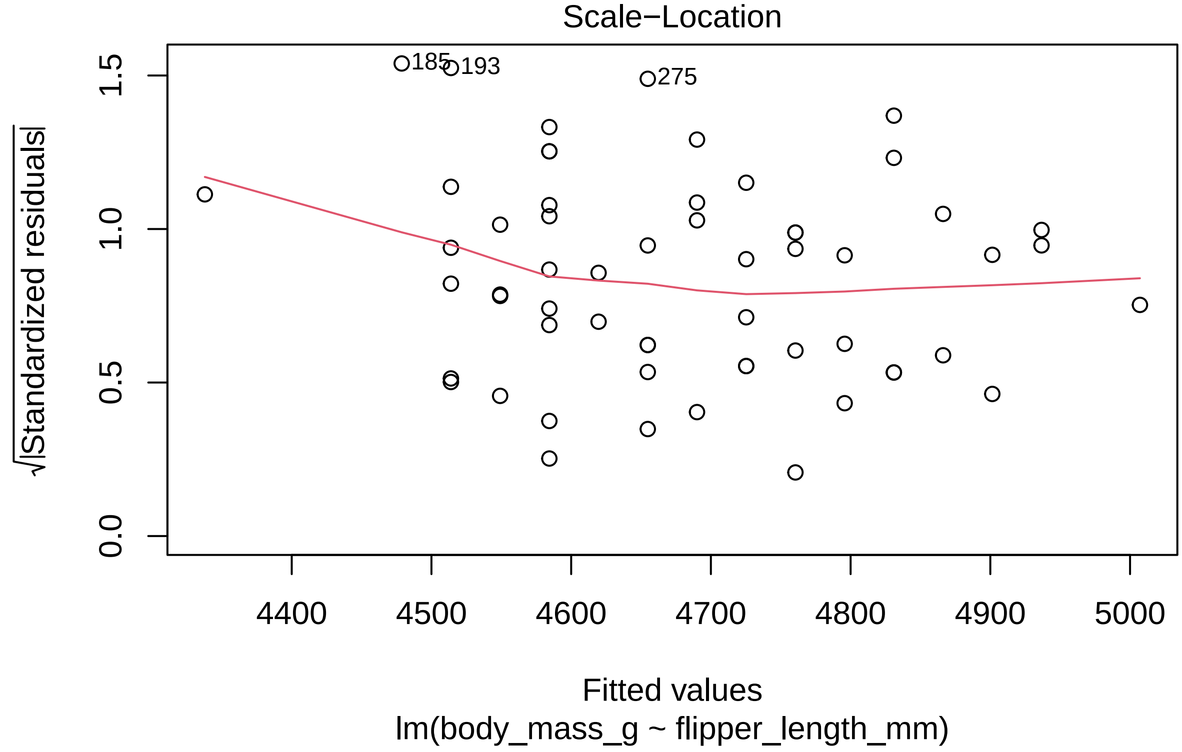
```
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -2812.895   1792.979  -1.569 0.122319
## flipper_length_mm    35.225      8.428   4.180 0.000103 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 248 on 56 degrees of freedom
## Multiple R-squared:  0.2378, Adjusted R-squared:  0.2242
## F-statistic: 17.47 on 1 and 56 DF,  p-value: 0.0001035
```

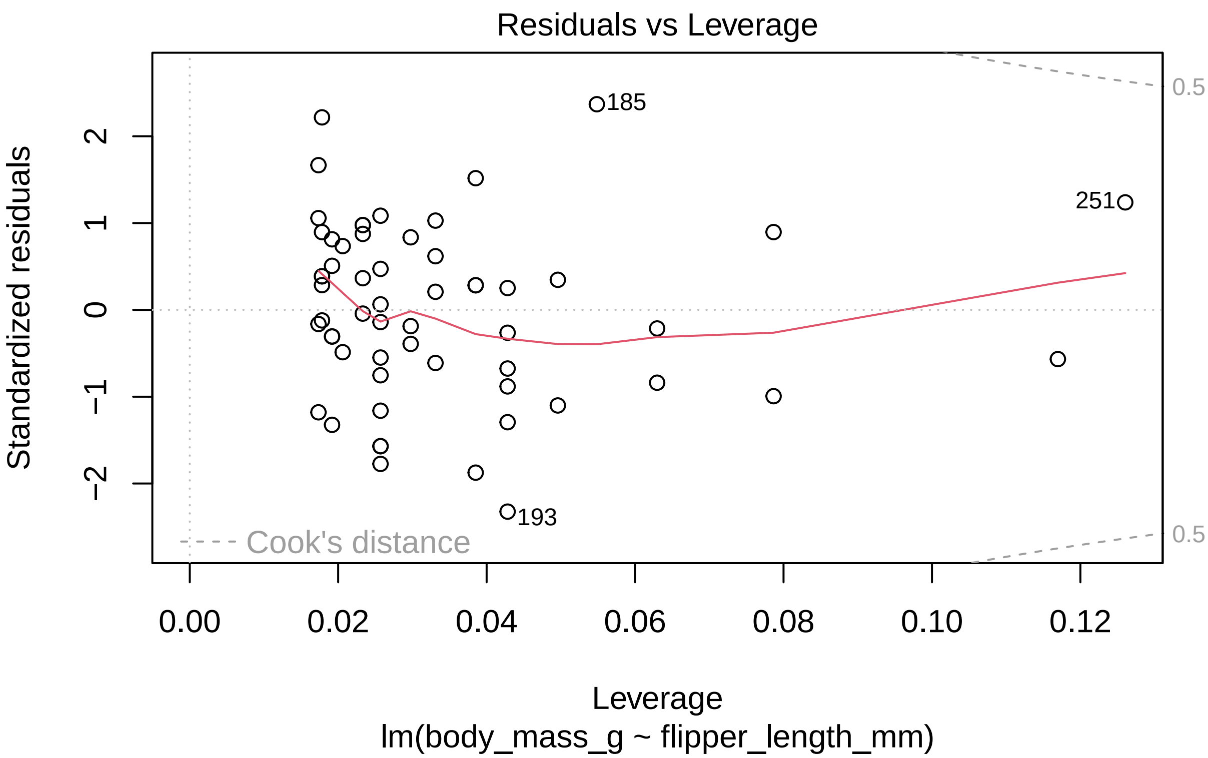
4.2.3.2 Assumption Checks Before making any conclusion, we need to check the residuals to ensure the assumptions are verified. This is based on the **model residuals**.

```
plot(mod)
```

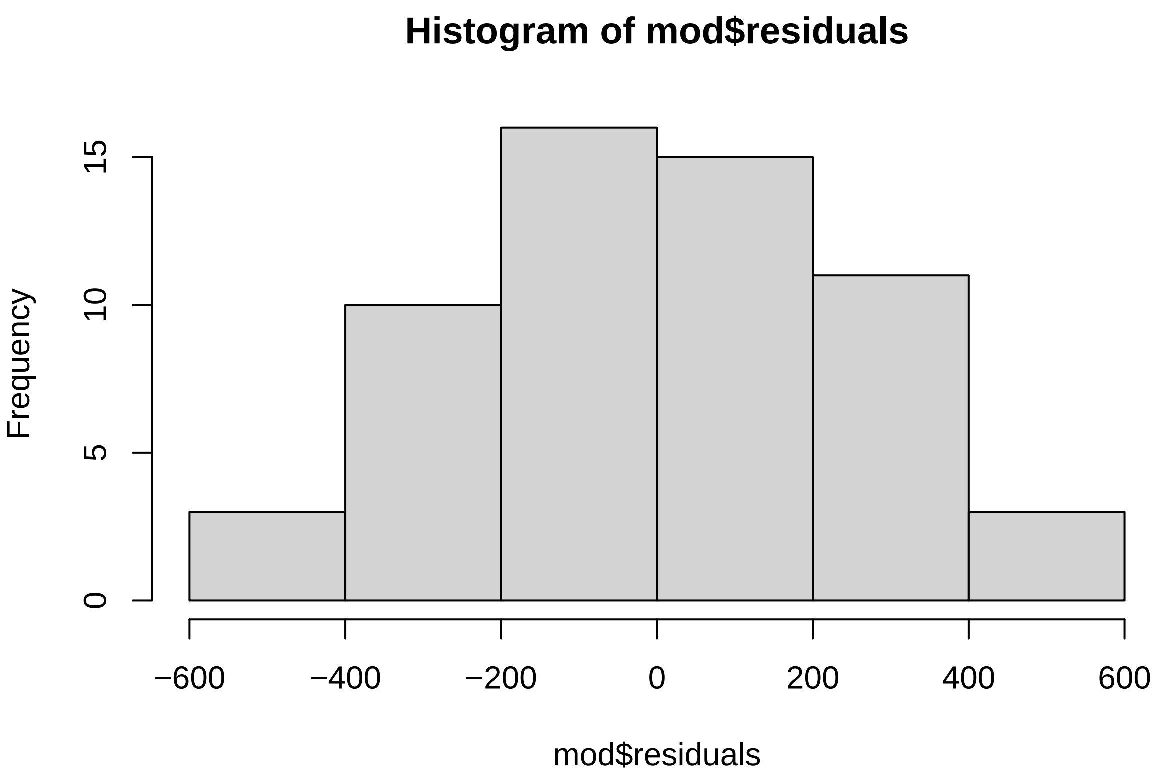



Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(body_mass_g ~ flipper_length_mm)
193
275
185
−2
−1
0
1
2


Scale−Location
√|Standardized residuals|
Fitted values
lm(body_mass_g ~ flipper_length_mm)
185
193
275
0.0
0.5
1.0
1.5
4400
4500
4600
4700
4800
4900
5000




```r
hist(mod$residuals)
```




```r
shapiro.test(mod$residuals)
```

```
##
##  Shapiro-Wilk normality test
```

```
##
## data:  mod$residuals
## W = 0.9907, p-value = 0.9365
```

```r
ncvTest(mod)
```

```
## Non-constant Variance Score Test
## Variance formula: ~ fitted.values
## Chisquare = 2.206033, Df = 1, p = 0.13747
```

The distribution of the residuals supports the normality and homoscedasticity assumptions.

4.2.3.3 Conclusion Based on a linear regression, we reject the null hypothesis. There is a significant association between flipper length and body weight ($p < 0.05$), with an effect size $\pm$ standard error of 35 ± 8 (for every additional mm, a penguin is expected to weight 35 more g). The R^2 correlation coefficient of 0.24, indicates weak association (a lot of the variability remains unexplained).

5 Associations Between Continuous and Catergorical Variables

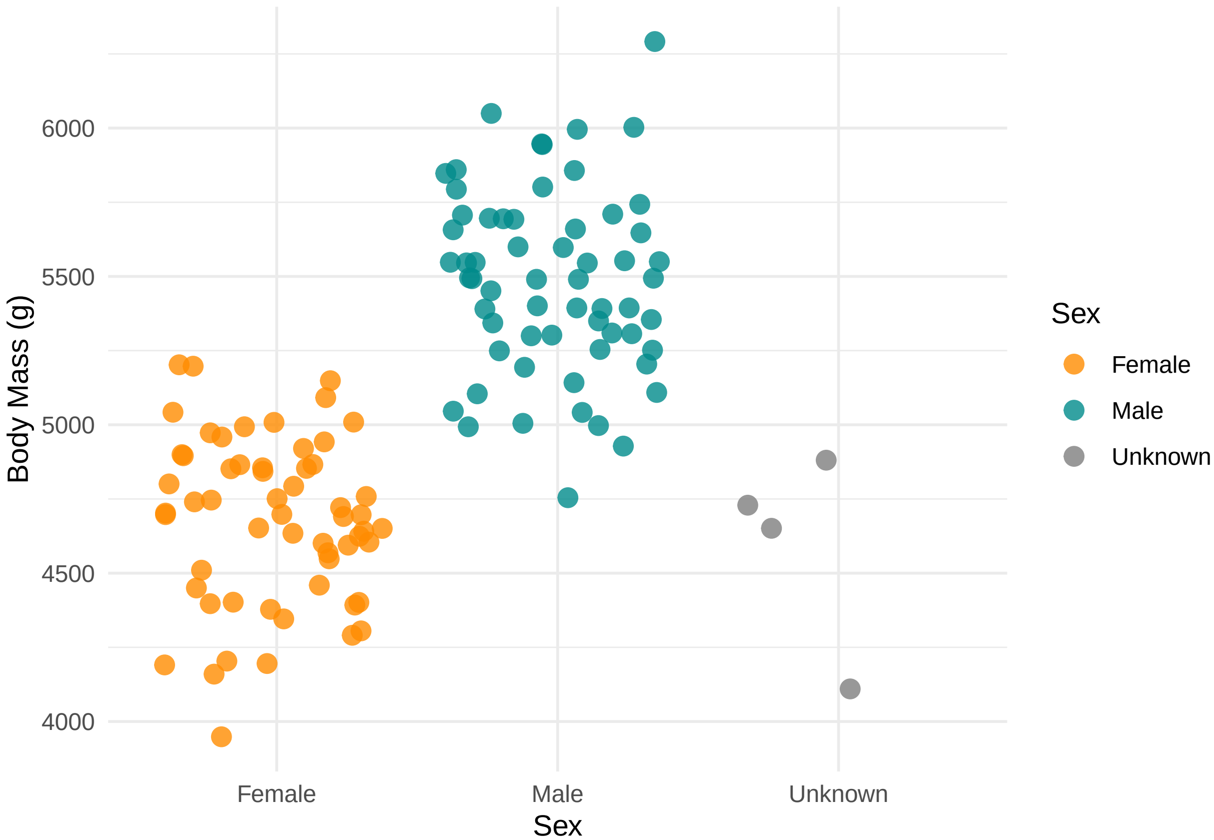
5.1 Subset

Let's now focus on one species to look at the association of weight with sex.

```r
dat_gen = subset(dat, species == "Gentoo")
```

5.2 Re-plot

```r
ggplot(dat_gen, aes(x = sex, y = body_mass_g, color = sex)) +
  geom_jitter(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                     name = "Sex",
                     labels = c("Female", "Male", "Unknown")) +
  labs(
    x = "Sex",
    y = "Body Mass (g)"
  ) +
  scale_x_discrete(labels = c("Female", "Male", "Unknown")) +
  theme_minimal()
```



6 Statistical Test

We saw before this relationship could be explored with a chi2 or an ANOVA, but they return limited information (notably no effect sizes). We can use linear regressions here too.

6.1 Assumptions

6.1.1 Independance

Observations from each population are independent of one another. This is likely the case (would be determined by the study design).

6.1.2 Normality

For a linear regression - *Response variable is normally distributed for each value of the explanatory variable.* This is easier to assess on the model residuals (after model fitting), although we can already guess on the scatter plot.

6.1.3 Homoscedasticity (regression)

Variance of the response variable is equal across each value of the explanatory variable. This is easier to assess on the model residuals (after model fitting), although we can already guess on the scatter plot. .

6.2 Test

- Null hypothesis: there is no association between sex and body weight.
- Alternative: there is an association between sex and body weight (i.e., variations in individual sex contributes to explain variations in individual weight).

6.2.1 Linear Regression Model

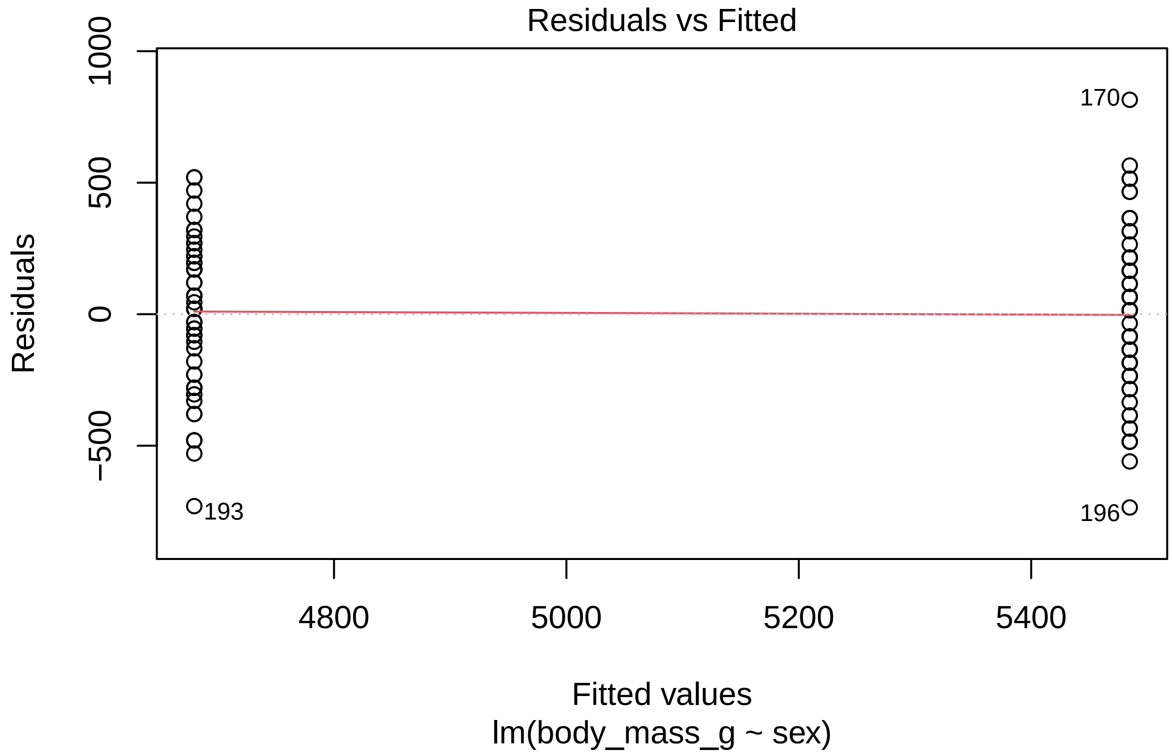
```r
# Fit model
mod = lm(body_mass_g ~ sex,
         data = dat_gen)

# Return outputs (including p-value)
summary(mod)
```

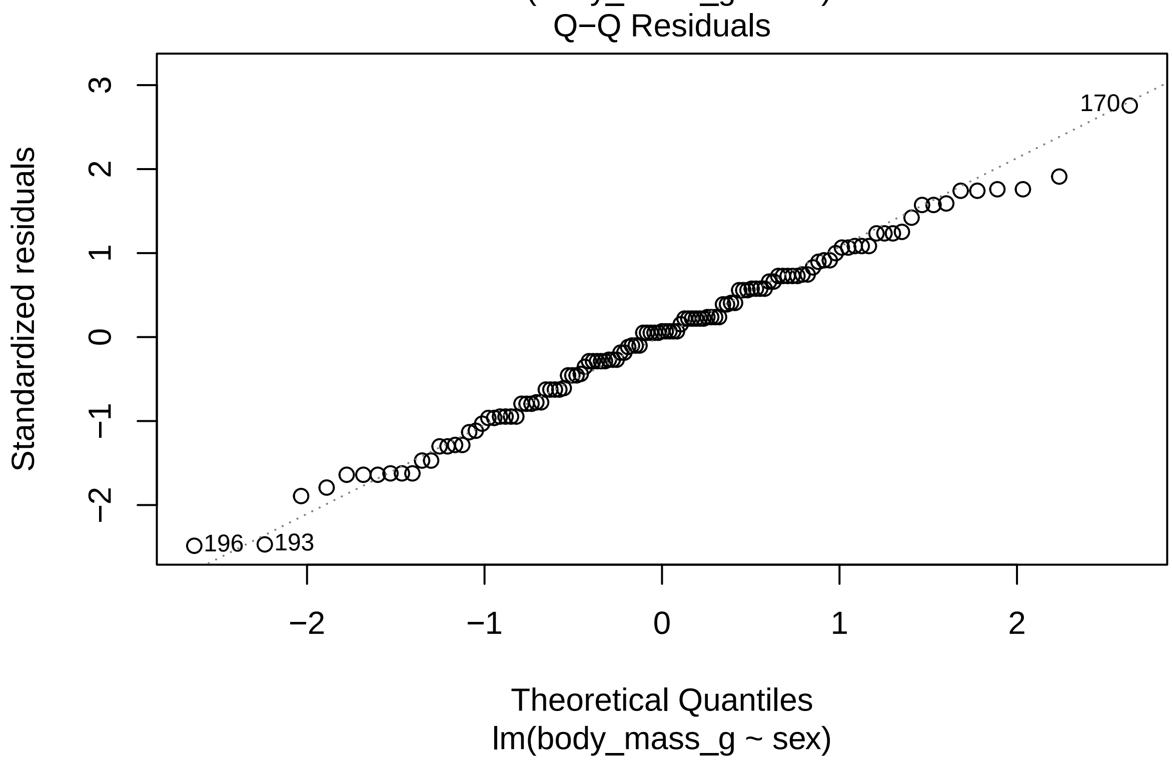
6.2.1.1 Model Fitting

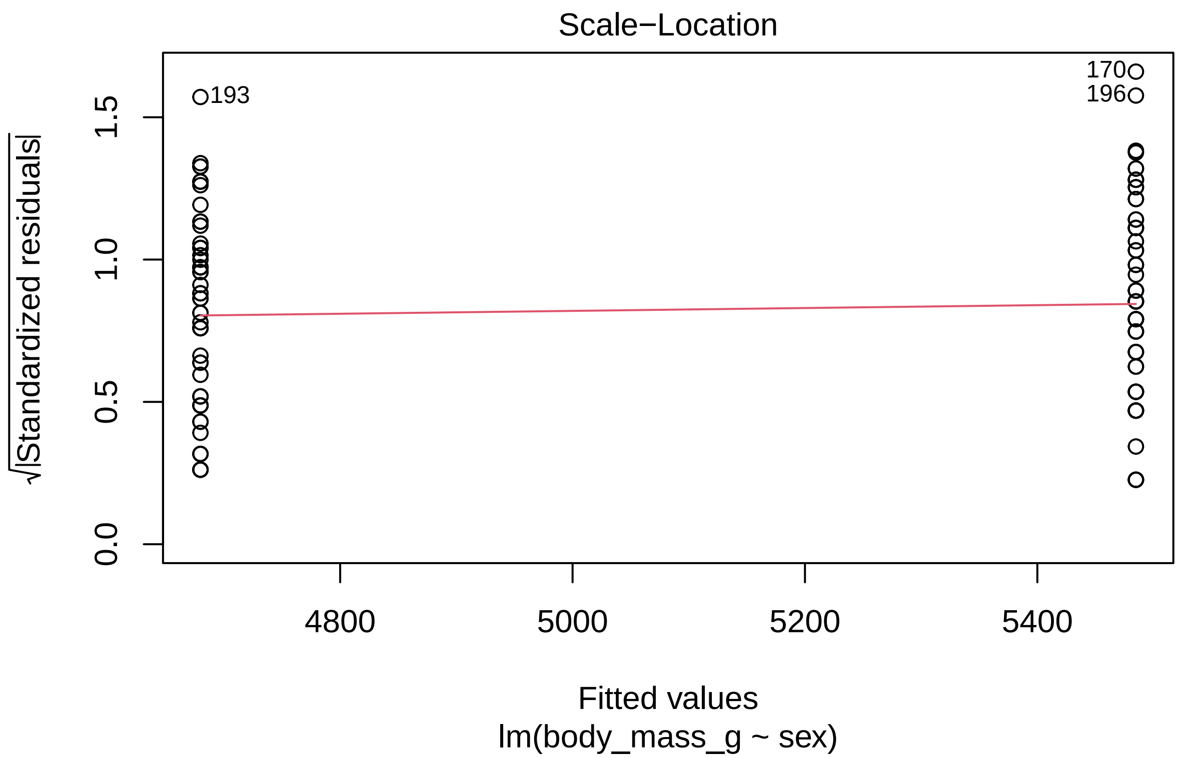
```
##
## Call:
## lm(formula = body_mass_g ~ sex, data = dat_gen)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -734.84 -207.29   20.26  215.16  815.16
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4679.74      39.15  119.52   <2e-16 ***
## sexmale       805.09      54.69   14.72   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 298.2 on 117 degrees of freedom
##   (5 observations deleted due to missingness)
## Multiple R-squared:  0.6494, Adjusted R-squared:  0.6464
## F-statistic: 216.7 on 1 and 117 DF,  p-value: < 2.2e-16
```

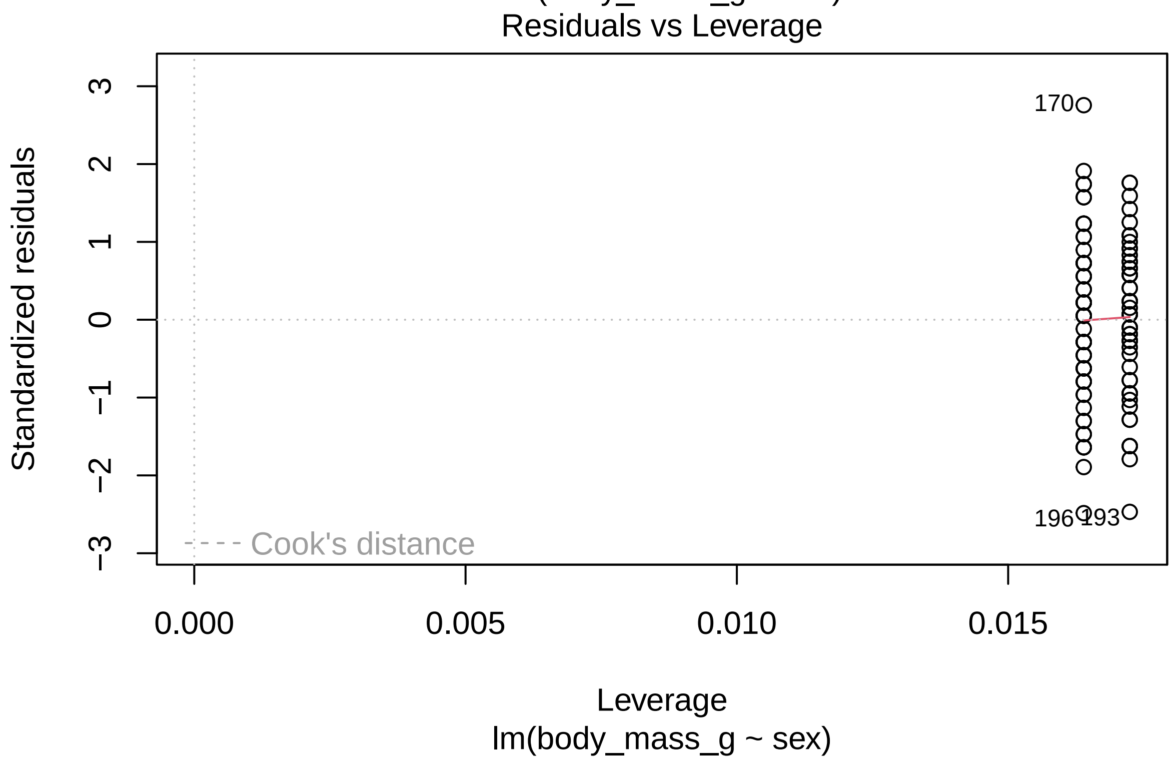
6.2.1.2 Assumption Checks Before making any conclusion, we need to check the residuals to ensure the assumptions are verified. This is based on the **model residuals**.

```r
plot(mod)
```

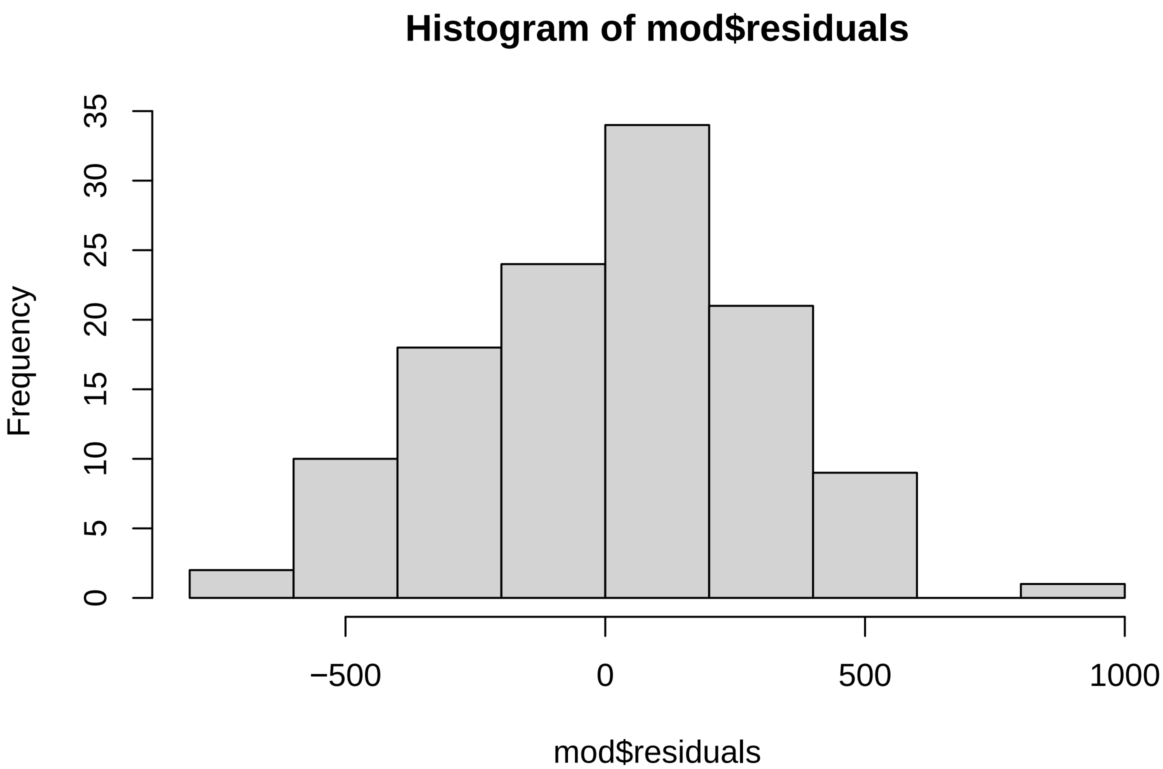

Residuals vs Fitted
Residuals
1000
500
0
−500
170
193
196
4800
5000
5200
5400
Fitted values
lm(body_mass_g ~ sex)


Q−Q Residuals
Standardized residuals
3
2
1
0
−1
−2
170
196
193
−2
−1
0
1
2
Theoretical Quantiles
lm(body_mass_g ~ sex)






```
hist(mod$residuals)
```




```
shapiro.test(mod$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  mod$residuals
## W = 0.99236, p-value = 0.7596
```

```
ncvTest(mod)
```

```
## Non-constant Variance Score Test
## Variance formula: ~ fitted.values
## Chisquare = 0.6686562, Df = 1, p = 0.41352
```

The distribution of the residuals supports the normality and homoscedasticity assumptions.

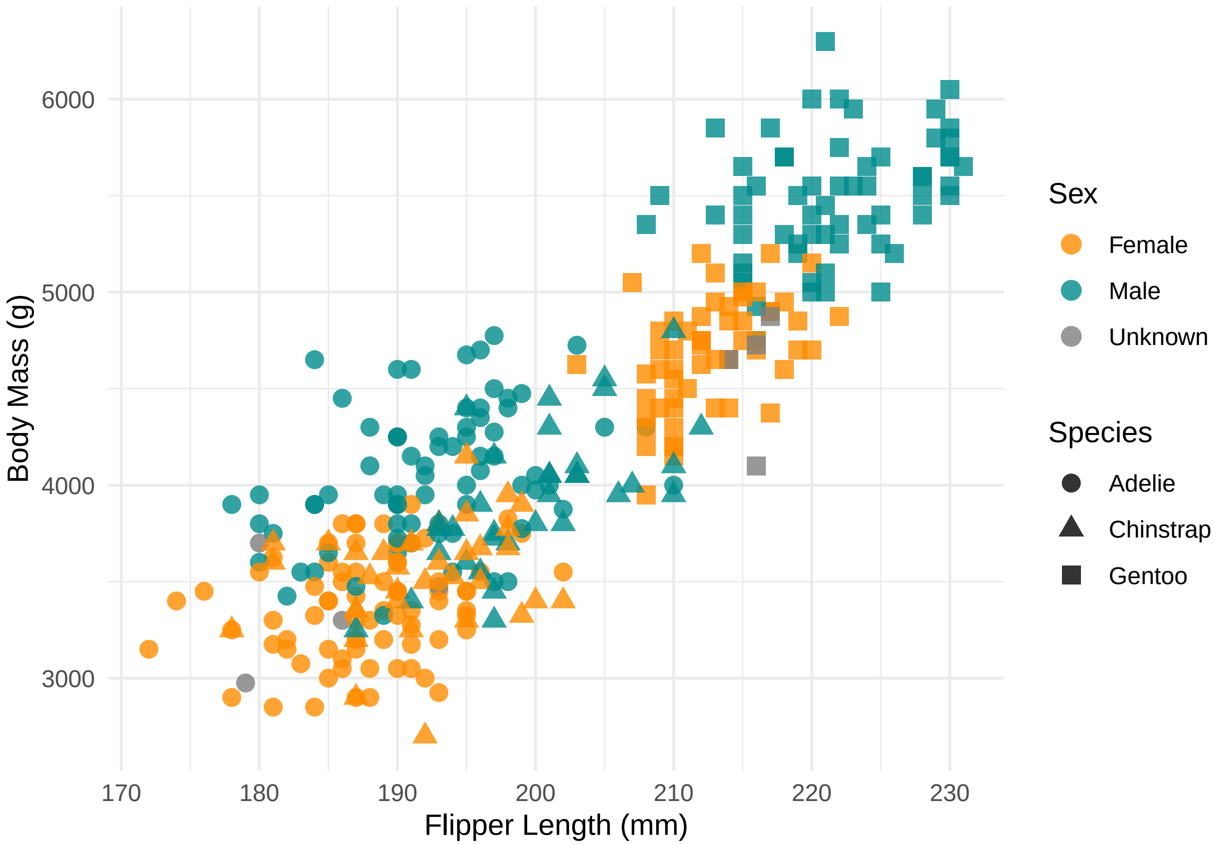
6.2.1.3 Conclusion Based on a linear regression, we reject the null hypothesis. There is a significant association between sex and body weight ($p < 0.05$), with an effect size ± standard error of 805 ± 55 (males are 805 g heavier than females). The R^2 correlation coefficient of 0.65, indicates moderate association (some of the variability remains unexplained).

7 Multivariate Models

Several explanatory variables can be included in regression models.

```
ggplot(dat, aes(x = flipper_length_mm, y = body_mass_g, color = sex, shape = species)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                     name = "Sex",
                     labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
```

```
                    name = "Species") +
  labs(
    x = "Flipper Length (mm)",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```




```
# Fit model
mod = lm(body_mass_g ~ flipper_length_mm + sex + species,
          data = dat)

# Return outputs (including p-value)
summary(mod)
```

```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm + sex + species,
##     data = dat)
##
## Residuals:
##    Min      1Q  Median      3Q     Max
## -721.8 -195.7    -5.9   198.6   873.7
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         -365.817    532.050  -0.688   0.4922
## flipper_length_mm     20.025      2.846   7.037 1.15e-11 ***
```

```
## sexmale            530.381     37.810  14.027  < 2e-16 ***
## speciesChinstrap   -87.634     46.347  -1.891   0.0595 .
## speciesGentoo      836.260     85.185   9.817  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 295.6 on 328 degrees of freedom
##   (11 observations deleted due to missingness)
## Multiple R-squared:  0.8669, Adjusted R-squared:  0.8653
## F-statistic:   534 on 4 and 328 DF,  p-value: < 2.2e-16
```

We can see that a large part of the variability is now explained! We will talk more about those models in the coming sessions…